Artificial Intelligence in Software Startups and Special Challenges in Least Developed Countries: A Case Study of Uganda

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***Abstract*—Artificial Intelligence (AI) is gradually transforming the landscape and operations of software startups by enabling innovation, improving decision making, and automating their business processes. However, software startups in the least developed countries (LDCs) like Uganda face a number of challenges that have hindered their abilities to adopt AI in their processes, products and services. In this study, we investigate the current state of AI in software engineering startups and explore challenges and opportunities of AI adoption in software startups in LDCs with a case study of Uganda. We used a mixed method approach that included interviews and online surveys involving 30 software startups operating in various sectors, including FinTech, EduTech, AgriTech, HealthTech, and E-commerce. Findings from our research highlight critical challenges affecting these startups in their pursuit of AI adoption such as limited access to funding, high cost of AI implementation, limited access to skills and technical expertise, lack of awareness and understanding of AI, and regulatory challenges. The study concludes by giving recommendations, best practices and future research for AI adoption in software startups in LDCs.**

***Index Terms*—Artificial Intelligence, Software Startups, Adoption, Least Developed Countries, Uganda**

I. Introduction

Software industry worldwide is experiencing a rapid growth driven by AI technologies like machine learning (ML), natural language processing (NLP), and computer vision. Software startups are leveraging AI for competitive advantage [1], [2], yet those in LDCs face structural and operational challenges that limit their capacity to harness these technologies. Uganda, classified as an LDC, provides a unique context to explore how software startups can overcome barriers to AI adoption. Despite the potential benefits of AI, software startups in Uganda struggle with challenges such as limited access to capital, inadequate infrastructure, and a shortage of skilled professionals. This paper seeks to explore these challenges and identify strategies to promote AI adoption in Ugandan software startups. This research contributes to the understanding of AI adoption in software startups within LDCs, providing insights for policymakers, investors, and entrepreneurs.

II. Overview of AI in Software Development

Artificial Intelligence (AI) is becoming a life changing force in the field of software engineering, significantly changing the software development process from designing, coding, testing and maintenance. As well as from improving code quality to automating multiple repetitive tasks, AI is helping software engineers to work more efficiently with ease and flexibility, hence minimizing errors and rapidly driving innovation. AI has further transformed the coding process by providing different tools that help software developers in writing and generating code more easily and efficiently [3], [4]. With the help of various AI-powered platforms such as GitHub Copilot and ChatGPT [5], developers can now utilise different machine learning algorithms to autocomplete code, suggest and recommend the relevant functions, and even generate boilerplate code. These tools use NLP models that have been trained on huge data repositories of public code to predict what a developer is most likely to write next.

Understanding user requirements is another important aspect of software engineering, that can impact the success of any software engineering project. AI technologies, more especially NLP, simplify the entire process of analysing large volumes of unstructured data, such as huge emails, user stories, and documentation [6]. These AI enabled requirement analysis tools can easily extract meaningful insights from these huge data sources, helping engineering teams to align software features with the user needs.

AI's integration into DevOps [7] is also changing how software engineering teams deploy, monitor, and maintain software systems. The AI enabled platforms can automate

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different routine maintenance tasks, optimize infrastructure, and even predict system failures when they are about to occur. Further more, by analysing different logs, metrics, and performance data, AI can alert teams to any potential issues and even give automated remediation, hence leading to more stable, resilient, and reliable systems.

AI's role in automated testing is another key aspect of AI in software development [8] . Traditional software testing methods have proved to be time consuming and prone to human error, but AI-powered testing tools can easily perform comprehensive and exhaustive testing within a short period of time. In various studies, [9], [10], researchers have highlighted several tools like Selenium and Testim, which leverage AI to create, execute, and maintain a series of test cases. By using AI/Machine learning algorithms to analyze past test cases and bug reports, these tools can be able to predict any potential problem areas and prioritize testing efforts, hence enhancing the software's reliability and performance [11].

In other works [12], [13], researchers also discuss how AI can facilitate continuous integration/continuous deployment by automating test execution, hence reducing the time to market for new software releases.

Beyond coding and software testing, AI also plays a critical role in project management. AI enabled project management tools like JIRA advanced roadmaps among other can help in optimizing resource allocation, predicting project timelines, and risk identification [14].

By analysing huge historical project data, AI can help in giving insights into potential bottlenecks and propose different mitigation strategies, thereby improving project efficiency. Furthermore, the predictive capabilities of AI can help in enhancing decision making by empowering project managers with data-driven insights, which helps in better planning, resource management, and delivery. To sum up, AI is fundamentally transforming the software development landscape by automating repetitive tasks, enhancing and improving code quality, and accelerating project timelines. While there are challenges and ethical considerations that need to be addressed, research and literature shows that, the benefits of adopting and integrating AI into the software development lifecycle are undeniable. As these AI technologies continue to evolve, we should expect even greater efficiencies and innovations, paving the way for a new era of software development.

III. A Global Perspective of AI in Software Engineering Startups

AI has become an important driver of innovation, enabling software startups to enhance their products, optimize operations, and gain a competitive edge. However, the pace and extent at which these startups adopt AI vary significantly across different regions, particularly between developed and least developed countries (LDCs).

A. Access to AI Infrastructure and Resources

Software engineering startups in developed regions such as North America, Western Europe, and parts of East Asia benefit from a well established AI infrastructure. They get access to high end technologies, robust and reliable cloud computing platforms, and high end computing resources [2], [15]. For example, the Silicon valley in the United States is a global leader in AI research, innovation and development, supported by technology giants such as Google, Microsoft, and Amazon, to mention but a few that offer AI as a Service (AIaaS) platforms (like the Google AI Platform, and Microsoft Azure AI). In Europe, countries like Germany, the UK, and France have made significant and strategic investments in AI research and innovation centers, creating a conducive environment for startups to experiment with latest AI technologies [16]. When it comes to Asia, countries like Japan and South Korea are well known for their strong emphasis on AI in manufacturing and industrial complexes, which provides software startups with more advanced and reliable AI tools and stable infrastructure [2].

On the other hand, software engineering startups in LDCs, particularly in regions such as sub-Saharan Africa and Uganda in particular, often struggle with challenges such as limited access to AI infrastructure, high costs of computing resources and AI infrastructure, like GPUs and high end servers, and poor internet connectivity that has hampered access to cloud based AI services, making it very difficult for startups to leverage and adopt AI tools in their processes [17].

B. Availability of Skilled Talent

Software startups in the developed regions have been blessed with access to a large pool of well trained and skilled AI professionals and practitioners, including data scientists, machine learning engineers, and AI researchers. This is not by chance but as a result of various factors that have contributed to the availability of skilled talent which include the presence world class education with top universities like MIT, Stanford, and ETH Zurich, producing highly skilled top graduates in AI and data science. Furthermore, developed countries have established vibrant AI Communities such as technology meetups and high-end conferences, which foster collaboration and knowledge sharing. Government support through strategic national AI strategies promote AI education, research, and talent development especially among the youth [18].

When it comes to LDCs, they still face significant challenges in building a skilled youthful AI workforce and this is mainly due to limited educational resources where only few universities in LDCs offer specialized AI or data science programs and low levels of core research and innovation output, hence leading to a shortage of local expertise especially in the area of AI and data science [19]. To make it worse, LDCs are faced with the challenge of brain drain where most talented individuals especially the youth often migrate to developed countries for better opportunities, leaving a high skills gap in their home countries. The talent gap is a critical barrier to AI adoption in software startups within LDCs, whereas developed countries benefit from a well-established pipeline of skilled AI professionals.

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C. Access to Funding and Investment

AI adoption in software engineering startups varies widely between the developed and the least developed countries, driven by differences in infrastructure, talent availability, funding, regulatory support, cultural readiness, among others. Software startups in developed countries are well positioned to leverage AI adoption in their processes due to their access to resources, skilled talent, and supportive ecosystems [16]. Conversely, startups in the LDCs face significant challenges that hinder their ability to adopt and scale AI technologies.

IV. Challenges of AI Adoption by Software Engineering Startups in LDCs

In LDCs, One of the most critical challenge faced by the software startups is the lack of and limited access to funding. AI projects are capital intensive, they need a substantial amount of financial resources to facilitate research, design and development, talent acquisition, and stable infrastructure. However, it has not been easy for these startups to secure the necessary and much needed capital/financial resources. According to The World Bank report [20], most investors in LDCs are typically more conservative, which biases them towards traditional industries over the emerging technologies like AI. This is because some of them perceive AI investment in LDCs as a venture with high risk and uncertainties hence, deterring investment.

Furthermore, in one of the research conducted by The International Finance Corporation (IFC) [21], they have pointed out that the availability of venture capital in LDCs is much lower as compared to the developed countries. The study goes ahead to highlight that startups in the Sub Saharan Africa get less than 1% of the global venture capital funding, hence, limiting their ability to invest in AI driven projects which are well known to be capital intensive. In another report by the United Nations Conference on Trade and Development (UNCTAD) [22], they state that government support for the tech startups in LDCs is very minimal as compared to the developed countries where grants and subsidies are much available for fostering AI research, LDCs lack similar financial incentives as their counterparts. This limited/lack of funding restricts software startups in LDCs from exploring and experimenting with AI technologies, hiring skilled personnel, and scaling their solutions to the next level. This financial constraint is one of the major reasons why AI adoption among software startups in these regions remains low [23].

Another challenge is the lack of reliable and stable infrastructure some of which includes, access to high speed internet, cloud computing, and reliable data centers, among others that are crucial for AI development. Unfortunately, the LDCs often lack the necessary infrastructure to support their AI initiatives. In one of the reports by the International Telecommunication Union (ITU) published in 2022, it states that only about 27% of the population in LDCs has access to the internet as compared to over 80% in the developed nations. This digital divide between the developed and LDCs significantly hampers the ability of these software startups to leverage different AI technologies, which often rely on cloud based platforms. A study by the African AI research institute in 2021 found out that access to advanced computing resources like GPUs and cloud services is very limited in LDCs. Forcing startups to rely on outdated hardware, which affects their ability to train and deploy complex AI models effectively. All this has led to slower innovation cycles hence limiting the competitive edge of these startups in the global tech ecosystem.

Regulatory Challenges is another critical factor affecting software startups in LDCs in adopting AI. Government regulatory environment plays an important role in facilitating the adoption of AI technologies. Unfortunately, most LDCs lack clear and supportive regulations for AI, which creates uncertainty for these startups. In the report published by the Global Partnership on Artificial Intelligence (GPAI) [24], most LDCs have not yet developed comprehensive AI strategies. This absence of a clear regulatory framework has left startups without a clear guidance on compliance and ethical AI deployment. This regulatory gap, totally discourages these software startups from adopting AI, especially for data intensive applications like machine learning and data analytics.

Regulatory challenges in LDCs have created a weak environment for software startups, limiting their ability to attract investments that might help them in scaling their operations.

V. Methodology

In this study, we explore the different opportunities and challenges of AI adaptation in software startups in LDCs with a case study of Uganda, we used a mixed method approach that included interviews and online surveys.

A. Research Context

Uganda's software startup ecosystem is gaining momentum as a vibrant hub for innovation in the East African region. Several startups have gained prominence by leveraging technology to solve local challenges. For example, SafeBoda, a mobile application that helps in connecting users (passengers) to motorcycle taxis commonly known as boda bodas for safe and efficient transport. SafeBoda has grown to become one of Uganda's most successful startups, expanding their operations to other countries like Nigeria and Kenya. Also, Yo! payments, specializing in mobile payments, provides a platform that integrates with mobile network operators, enabling startups to offer seamless payment solutions. ClinicPesa, another healthtech startup offering payments and loans platform for medical expenses. It empowers users to save for healthcare, hence, addressing the financial barriers to accessing medical services.

B. Key Players in the ecosystem

Technology hubs and incubators have proved to play an important role in the nurturing of the software startups in Uganda. These organizations provide startups with access to mentorship, funding, and networks essential for their growth. For example, Outbox Hub is one of Uganda's leading tech incubators that was established in 2012 and it is well known

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for offering startups with co-working spaces, training needs, and different funding opportunities. It has supported more than 200 software startups.

Another key player is the Innovation Village that supports startups through mentorship, training programs, and access to venture capital. It focuses on scaling tech startups by fostering partnerships with local and international organizations. There is also Hive Colab that has been instrumental in giving software startups with access to resources and collaboration opportunities. It mainly focuses on sectors such as fintech, healthtech, and agritech . Another key player is the government of Uganda, which has increasingly tried to support the growth of its tech ecosystem through policies and initiatives aimed at fostering innovation. For example, in 2017 the government launched the National ICT Initiatives Support Program (NIISP) to give grants to local software developers and startups to promote homegrown digital solutions. This program has been pivotal in boosting the capacity of Uganda's software sector. The government of Uganda has gone ahead to set up the National ICT Innovation Hub to help in fostering homegrown startups in digital solutions and promoting digital employment throughout the country. The HUB has supported over 50 startups.

C. Data Collection

In order to achieve our research objective, we collected data from 30 software startups operating in various sectors, including FinTech, EduTech, AgriTech, HealthTech, and e-commerce.

Through the ICT Association of Uganda, Data Science Association forum, and social contacts, we were able to get contact details of different key stakeholders such as CEO's CTO's, software engineers, and directors/founders of the startups operating in Uganda. Using these contacts, we shared the survey instrument via Google forms with them. We gave them two weeks to respond to the survey. All respondents responded to the survey instruments within the set time frame. At the end of the two weeks, we extracted the data for analysis.

VI. Discussion of Findings

A. General Findings

1) Demographics of surveyed Startups: Findings show that of the 30 surveyed startups, 33.3% have been in operation for over 5 years, 13.3% for 3-4 years, 26.7% for 1-2 years and 26.7% for less than a year. In addition, the employment capacity is low with 73.3% having 1-10 employees, 20% having 11-20 employees, and 6.7% having above 50 employees.

2) Current sectors of operation of surveyed Startups: The survey of the 30 software startups in Uganda revealed a growing trend in the adoption of AI technologies. The majority of these startups were focused on EduTech, accounting for 38.5%, followed by HealthTech at 23.1%. Fig. 1 illustrates the distribution of software startups across these sectors based on the survey results.

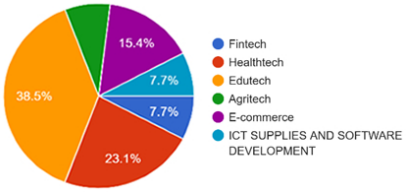


Fig. 1. Software Startups in Uganda

3) Current State of AI Adoption in Ugandan Startups: Notably, 76.9% of these startups are currently integrating AI into their processes, products or services, showcasing a significant commitment to leveraging advanced technologies to enhance their offerings. Fig. 2 illustrates the adoption of AI technologies based on the survey results.

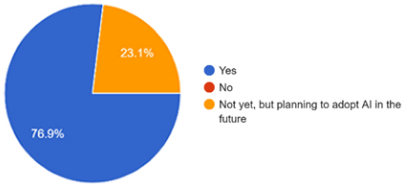


Fig. 2. Adoption of AI Technologies.

Among the AI technologies employed, Machine Learning is the most prevalent, with 61.5% of the startups utilizing it for various applications, such as predictive analytics and automation. Furthermore, Natural Language Processing (NLP) is used by 30.8% of the startups, primarily for text analysis and language based solutions. Interestingly, chatbots are also widely adopted, with 46.2% of the startups implementing them to improve customer engagement and support.

On the more specialized end, Computer Vision is also used by 15.4% of the startups, enabling computing capabilities like image recognition, and video analytics. Meanwhile, the Generative AI, which seems to be a relatively emerging technology, is currently being explored by 7.7% of these software startups, often for content creation and innovative product features. This data indicates a strong inclination towards AI adoption among Ugandan startups, particularly in leveraging AI for automation, customer interaction, and advanced analytics. Fig. 3 illustrates the AI technologies being used by startups based on the survey results.

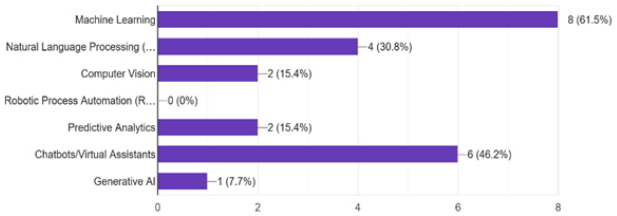


Fig. 3. AI Technologies Being Used by Startups

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B. Key Opportunities for AI in Uganda

The survey further highlights significant opportunities and growth potential for the local tech ecosystem, particularly in the area of AI adoption. The findings reveal several key drivers behind the adoption of AI among these startups, which are fueling their expansion and competitiveness in the market. Fig. 4 illustrates the key drivers for adoption of AI among startups based on the survey results.

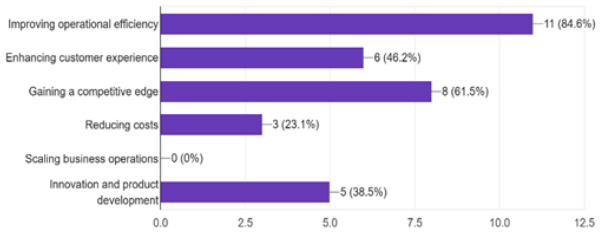


Fig. 4. Key Drivers for Adoption of AI Among Startups

The most compelling motivation for embracing AI is improving operational efficiency, cited by 84.6% of the startups. By automating routine tasks, optimizing workflows, and streamlining processes, startups are able to achieve higher productivity levels and reduce operational bottlenecks, ultimately accelerating their growth trajectory. Another major driver is the desire to gain a competitive edge, with 61.5% of the startups leveraging AI to differentiate themselves in an increasingly crowded market. By deploying AI-powered solutions, these startups are able to offer unique features, enhance decision-making, and respond more agilely to market changes, thereby positioning themselves ahead of competitors. Enhancing customer experience is also a priority, with 46.2% of startups adopting AI to improve customer interactions, personalize services, and deliver more responsive support. This focus on customer-centric innovation not only fosters brand loyalty but also helps startups capture a larger market share. Moreover, 38.5% of the startups are utilizing AI for innovation and product development, enabling them to create new solutions, optimize existing products, and explore untapped markets. AI-driven insights are empowering these startups to innovate faster and adapt to evolving customer needs.

Finally, reducing costs is a key factor for 23.1% of the startups, where AI is being used to lower operational expenses, optimize resource allocation, and drive cost efficiencies. This financial flexibility allows startups to reinvest in growth initiatives and scale their businesses more sustainably.

Overall, these drivers illustrate the vast opportunities for software startups in Uganda to harness AI for growth, efficiency, and innovation. As AI adoption continues to rise, these startups are well-positioned to thrive in the dynamic tech landscape, contributing to the broader digital transformation of the country.

C. Challenges Faced by Software Startups

The adoption of AI in Ugandan software startups presents significant potential for innovation and growth; however, it also comes with a set of challenges that hinder its widespread implementation. Based on a survey conducted among 30 software startups in Uganda, several key barriers to AI adoption have been identified, shedding light on the obstacles these startups face in leveraging AI technologies. Fig. 5 illustrates the key barriers to AI adoption among startups based on the survey results.

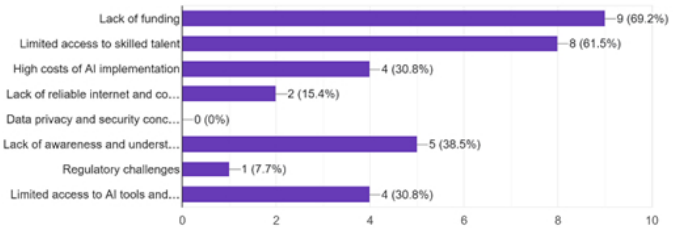


Fig. 5. Key Barriers to AI adoption Among Startups

The most prominent challenge is a lack of funding, cited by 69.2% of the surveyed startups. AI projects often require substantial financial investment, from acquiring advanced technologies to hiring skilled professionals. Without adequate funding, many startups struggle to kickstart or sustain AI initiatives, limiting their ability to scale and innovate. Closely following this is the limited access to skilled talent, reported by 61.5% of the startups. The shortage of professionals with expertise in AI, machine learning, and data science is a significant barrier, making it difficult for startups to build competent AI teams. This talent gap not only slows down the adoption process but also affects the quality and effectiveness of AI solutions developed.

Another major challenge is the lack of awareness and understanding of AI, affecting 38.5% of the startups. Many entrepreneurs and decision-makers lack the necessary knowledge of AI's potential benefits and applications, which can lead to reluctance in adopting these technologies. This gap in understanding can result in missed opportunities for leveraging AI to drive business growth.

The high cost of AI implementation is a barrier for 30.8% of the startups. Deploying AI systems often involves substantial expenses related to software, hardware, and infrastructure, making it a prohibitive option for many early-stage startups with limited budgets. Similarly, limited access to AI tools and platforms is a concern for 30.8% of the respondents. Startups often face difficulties in accessing the latest AI tools, frameworks, and platforms, which can limit their ability to experiment with and deploy AI solutions effectively.

While regulatory challenges were reported by only 7.7% of the startups, navigating the legal landscape around AI adoption can still pose complications, especially in areas related to data usage and compliance. This can deter startups from fully embracing AI technologies due to uncertainties around regulatory compliance. Interestingly, lack of reliable internet and computing infrastructure was highlighted by 15.4% of the startups, indicating that basic infrastructure remains a challenge in some regions. Reliable internet connectivity and

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access to cloud computing resources are critical for AI operations, and their absence can significantly hinder AI adoption.

Lastly, data privacy and security concerns were not cited as a barrier by any of the respondents, with 0% indicating this as a challenge. This suggests that either data privacy is not currently a primary focus for these startups, or they have found ways to manage these concerns effectively.

These findings underscore the need for targeted interventions to address the challenges faced by Ugandan startups in adopting AI. By addressing funding gaps, investing in talent development, and improving access to AI tools and infrastructure, there is significant potential to accelerate AI adoption and drive growth in the Ugandan tech ecosystem.

VII. Conclusion

This study has provided very important insights into the domain of AI adoption among Ugandan software startups, while focusing on the their current state, challenges, and potential pathways for growth. The findings from the research has identified a promising level of AI adoption, with most of the startups leveraging technologies like machine learning, NLP, and chatbots, among others to enhance operational efficiency, gain a competitive edge, and drive customer engagement. These opportunities highlight the growing role that AI can play in transformation of Uganda's tech ecosystem. However, the findings from the study also uncovered significant challenges that are hindering AI adoption, some of them include, limited funding, scarcity of skilled and talented personel in the field of AI and data science, high implementation costs, and limited access to advanced AI tools. These barriers underscore the need for targeted interventions to support startups in accessing both the financial and technical resources required for AI integration.

To address these challenges, we recommend initiatives such as increase in funding and government incentives where governments and private sector key players can design policies and comprehensive frameworks for technological funding and support of these startup, establish AI training programs, and improve access to affordable AI tools and infrastructure. These actions would strengthen the AI ecosystem with the startups hence, enabling more startups to innovate and scale effectively.

Future research should focus on exploring AI specific policy frameworks, long-term impacts of AI adoption on the economic growth of the country, and ways to foster public private partnerships that can support AI driven development in the least developed countries. This study contributes to the broader understanding of AI's role in fostering innovation in the developing countries and it serves as a foundation for future research and policy initiatives aimed at overcoming AI adoption barriers among software startups in Uganda and similar contexts.

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